

The validation methodology is based on the classification of the detected components into three critical categories, as shown in the figure. First, True Positives (TP) correspond to those elements present in the diagram that have been faithfully reproduced in the code. Second, False Positives (FP) group together the components that the model has generated incorrectly or arbitrarily, without any prior reference existing in the original design. Finally, False Negatives (FN) identify those elements omitted by the tool despite being part of the diagram's technical specification.

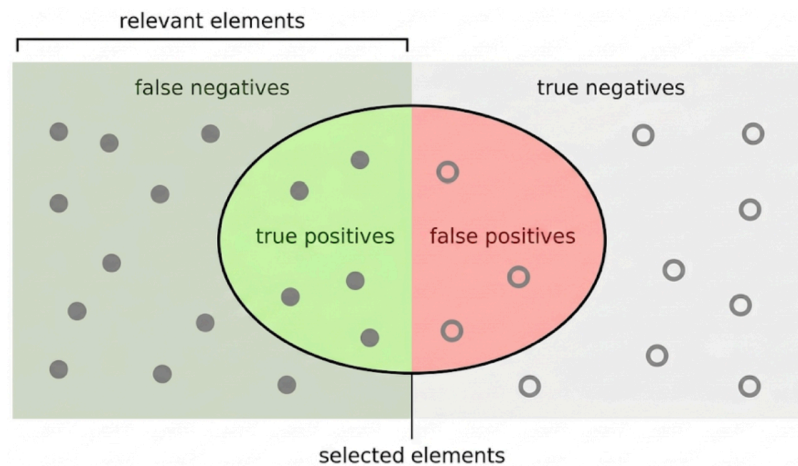


Figure. Diagram of true and false positives and negatives.

Because the impact of an omission or invention varies according to the nature of the component, a system of importance weights (W) has been designed to weight the technical impact of each success or error in the overall calculation:

Architectural element	Weight (W)	Technical justification
Classes / Entities	10	They constitute the structural foundation. Their absence compromises the integrity of the system.
Relationships	8	They define the hierarchy and the fundamental logical interactions.
Methods / Functions	5	They represent the expected behavior and operational logic.
Attributes / Properties	3	They define the state of the objects, being less critical than the class structure.
Packages	2	They provide organization to the code without directly affecting the functional logic.

Table. Weighting system for each architectural element for the validation of GENESIS.

To quantify the performance of GENESIS, the precision and recall metrics adapted through weight-based weighting are applied. Weighted Precision (Pw) evaluates the reliability of the system, determining what proportion of the generated code is accurate with respect to the original:

$$P_w = \frac{\sum VP \times W}{\sum (VP + FP) \times W}$$

Weighted Recall (Rw), or Recall, measures the tool's ability to retrieve all the relevant information from the design:

$$R_w = \frac{\sum VP \times W}{\sum (VP + FN) \times W}$$

Finally, the F1-Score is used as a metric of the harmonic balance between both variables. This consolidated value makes it possible to determine the platform's overall accuracy percentage, penalizing significant deviations in either of the previous metrics in order to provide a realistic view of the system's practical usefulness in a real engineering environment.

$$F1 = 2 \times \frac{P_w \times R_w}{P_w + R_w}$$

It is essential to emphasize that the validation results are intrinsically linked to the performance of the LLMs used. For this validation, qwen3-vl (8b) and qwen2.5-coder (14b) were used. It is also important to emphasize that the results can be improved by making manual adjustments to the PlantUML interpreted by the LLM or by requesting corrections to the final code.

Results and Quantitative Validation

The following table presents a detailed breakdown of the architectural components identified in the case study, distinguishing the elements classified as True Positives from those corresponding to False Positives or False Negatives:

Architectural element	VP	FP	FN
Classes / Entities	6	0	0
Relationships	3	0	2
Methods / Functions	18	0	0
Attributes / Properties	21	0	0
Packages	0	0	0
Precisión (Pw)	100%		
Recall (Rw)	93,68%		
F1-Score	96,74%		

Table. Empirical quantitative validation matrix for GENESIS.